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To: The Editor-in-Chief / Editorial Board
Subject: Submission of Original Research Manuscript

Dear Editor-in-Chief and Esteemed Members of the Editorial Board,

I am pleased to submit our original research manuscript entitled:

> **"MFHOI-Miner: Direct In-Process Mining of Maximal Frequent High-Occupancy Itemsets via Closure Leaps and Dual-Boundary Geometry"**

for consideration for publication as a regular research article in your prestigious journal.

Research Context and Problem Formulation Frequent Itemset Mining (FIM) and Maximal Frequent Itemset (MFI) mining are foundational paradigms in data engineering, pattern discovery, and knowledge extraction. While MFI mining effectively addresses the combinatorial explosion of patterns by identifying the positive boundary of frequent itemsets, it ignores the internal transaction completeness. High-Occupancy Itemset (HOI) mining was introduced to identify patterns that dominate the volume of transactions in which they appear. However, because occupancy is strictly *non-monotonic*—extending an itemset can arbitrarily increase or decrease its occupancy ratio—traditional Apriori/anti-monotonic pruning strategies fail completely.

Consequently, existing methodologies rely on either:

1. Mining all MFIs first and applying post-hoc occupancy filtering, which we theoretically and empirically prove can discard up to **94.2%** of valid maximal high-occupancy patterns due to non-monotonic occupancy dilution; or
2. Mining all frequent high-occupancy itemsets ($\mathcal{FHO}$) and executing a pairwise quadratic superset sweep, which suffers from catastrophic computational and memory overhead.

Core Contributions and Novelty In this paper, we resolve this long-standing algorithmic dilemma by formulating the **Dual-Boundary Geometry** of high-occupancy patterns and introducing **MFHOI-Miner**, the first exact, direct in-process algorithm for mining Maximal Frequent High-Occupancy Itemsets ($\mathcal{MFHOI}$). Key technical contributions include:

- **Galois Lattice Grounding (Theorem 1):** We prove that every maximal frequent high-occupancy itemset is strictly support-closed ($\mathcal{MFHOI} \subseteq \mathcal{FCI}$). This establishes that the true underlying search space is the compact Galois closure lattice rather than the exponential powerset.
- **Occupancy Suffocation Point (K^*, Theorem 2):** We establish an a-priori global depth ceiling directly derived from the database transaction length distribution, effectively bounding the maximum cardinality of any potential candidate before search traversal begins.
- **Closure Leap Pruning and Dual-Boundary Traversal:** Combining Head-Union-Tail (HUT) superset lookahead with Galois closure leaps, the algorithm skips entire subtrees

and equivalence classes in $O(1)$ 64-bit bitset operations without generating intermediate unclosed subsets.

- **Ternary State-Tracking and MFHO-Trie:** We devise a ternary state classification scheme to eliminate false-pruning anomalies during depth-first backtracking, coupled with an inverted rare-item prefix tree (MFHO-Trie) for ultra-fast superset containment tests.
- **Comprehensive Empirical Validation:** Benchmarking across 13 diverse real-world and synthetic datasets demonstrates that MFHOI-Miner prunes up to **98.5%** of search nodes directly in-process, accelerates mining throughput by several orders of magnitude over multi-stage baselines, and achieves extraordinary pattern compression ($|\mathcal{FI}| \gg |\mathcal{FHO}| \gg |\mathcal{FCI}| \gg |\mathcal{MFHOI}|$).

Significance and Relevance to the Journal This work bridges the theoretical gap between Galois concept lattices, non-monotonic constraint optimization, and high-performance bitwise pattern mining. Given the journal's focus on scalable data mining algorithms, database theory, advanced pattern discovery, and knowledge engineering, we believe this manuscript will be of strong interest to your readership and will serve as a foundational benchmark for non-monotonic maximal pattern discovery.

Declarations In accordance with ethical publishing standards, I confirm that:

- This manuscript represents entirely original research and has not been published previously.
- It is not under consideration for publication elsewhere, in whole or in part.
- All data sources and benchmarks utilized are standard, publicly available scientific repositories (FIMI and UCI).
- The author has approved the final manuscript and agrees with its submission.
- There are no conflicts of interest to disclose.

Thank you very much for your time, consideration, and coordination of the peer-review process. I look forward to hearing from you.

Sincerely,

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